

A Multi-sector Analysis of Current and Future Synergy of AI and Drones for Tunisia

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This case study investigates the transformative potential of integrating Unmanned Aerial Vehicles (UAVs) and Artificial Intelligence (AI) to address national priorities in Tunisia. While the use of UAVs for applications like precision agriculture, digital heritage preservation, and infrastructure inspection is well-established in developed nations, this research examines their practical integration within Tunisia's unique socio-economic context. The study analyzes applications across agriculture, cultural heritage, and infrastructure, demonstrating how this synergy enables data-driven approaches to challenges such as water scarcity and conservation. The novelty of this work, however, lies in its multi-sector comparative framework that analyzes AI-UAV adoption not as a mere technological upgrade but as a process of contextual appropriation. By systematically contrasting Tunisia's pragmatic, cost-driven model with Japan's automation-focused approach, a key finding is that the technological divergence represents a strategic adaptation to local constraints (e.g., water scarcity vs. labor shortages) rather than a simple development gap. This analysis provides a novel model for tailoring technology to resource-constrained economies, offering a transferable framework for policymakers and practitioners.

Key Words : *Agriculture, Archeology, Artificial Intelligence, Tunisia, Unmanned Aerial Vehicle*

1. INTRODUCTION

Drones have emerged as a transformative tool in Tunisian agricultural sector, with their implementation centered on a critical goal: enhancing agricultural efficiency and sustainability through precision farming practices. In this context, UAVs (Unmanned Aerial Vehicles) play multiple practical roles – they conduct crop inspections, evaluate soil conditions, manage irrigation systems, and measure crop yields. By leveraging this technology, Tunisian farmers can optimize the use of limited resources like water and fertilizer, detect crop diseases at an early stage, and reduce reliance on manual labor – addressing pressing challenges of water scarcity and climate variability in the region.

The drones used in Tunisia are tailored to diverse mission needs, with research institutions and private companies primarily adopting two types: multirotor and fixed-wing UAVs. Common models include the DJI Mavic, DJI Phantom, DJI Matrice, DJI AGRAS T and SenseFly eBee X. These drones are characterized by lightweight designs and user-friendly operation, and they support a range of sensors – such as RGB, multispectral, thermal, and

LiDAR (Light Detection and Ranging) cameras – to capture detailed agricultural data. For instance, using RGB and multispectral sensors for tasks like crop mapping and soil analysis.

To further unlock the value of drone data, Tunisia has used AI to process the data from UAV. AI enables automated data processing: its algorithms analyze aerial images to automatically detect crop diseases, identify weeds, predict yields, and classify soil types. This AI-UAV combination directly addresses following major agricultural pain points in Tunisia – small farm sizes, labor shortages, and insufficient water resources – by generating real-time data that empowers farmers to make faster, more informed decisions.

When compared to Japan, a global leader in smart agriculture¹⁾⁻¹⁰⁾, Tunisia's adoption of AI-based UAV systems is still in its early stages. Japan utilizes advanced UAVs equipped with AI-powered systems and high-end sensors for tasks like automated spraying and precision mapping. In contrast, Tunisia focuses on cost-effective and operationally flexible electric drones, which are well-suited to its dry environments and primarily used for monitoring and data collection.

Table 1 Purpose of using drones.

Application Area	Purpose	Benefits
Crop monitoring	Detect crop stress and disease	Improve yield, reduce losses
Soil analysis	Assess soil texture and moisture	Optimize irrigation and fertilization
Irrigation management	Monitor water distribution	Save water, increase efficiency
Yield prediction	Estimate productivity levels	Support planning and marketing

Table 2 Type of drones possibly using in Tunisia derived from the practical business cases, which were mainly around DJI Phantom, Mavic, AGRAS T series (samples as shown, DJI Phantom 4 RTK, DJI Matrice 300 RTK and senseFly eBee X).

Model	Type	Weight	Flight Time	Sensors	Main use
DJI Phantom 4 RTK	Multicopter	1.4 kg	30 min	RGB, multispectral	Crop mapping, soil health
DJI Matrice 300 RTK	Multicopter	6.3 kg	55 min	RGB, thermal, LiDAR	Large-field surveys, irrigation analysis
senseFly eBee X	Fixed-wing	1.6 kg	90 min	RGB, multispectral	Long-range mapping, yield estimation

Table 3 The implementation of AI technology with UAV systems in Tunisia.

Application	AI Technique	Purpose	Outcome
Disease detection	CNN (Convolutional Neural Networks)	Identify olive and cereal diseases	Early diagnosis, lower pesticide use
Weed detection	Image segmentation	Differentiate crops from weeds	Efficient spraying
Yield prediction	Regression models	Estimate productivity	Better resource planning
Water stress detection	NDVI analysis + ML	Identify drought zones	Optimize irrigation scheduling

Table 4 Comparison Between Tunisia and Japan in AI-Based UAV Applications.

Feature	Tunisia	Japan
Detection targets	Crops (olives, wheat, dates)	Rice, vegetables, fruits
Brands	DJI, SenseFly	Yamaha, DJI, Fujitsu
Drone size	Small to medium	Medium to large
Sensors	RGB, multispectral, thermal	RGB, hyperspectral, LiDAR
Energy source	Electric battery	Electric & solar-powered
AI integration	Experimental (research level)	Advanced (commercial scale)

2. PURPOSE AND OBJECTIVES

This article aims to analyze the current and future synergy of Artificial Intelligence (AI) and Unmanned Aerial Vehicles (UAVs) as a transformative tool for addressing Tunisia's national priorities. Its primary objectives are to investigate the practical integration of these technologies across key sectors – namely agriculture, cultural heritage, and infrastructure – by examining application purposes,

technology types, and AI's role in data processing. Furthermore, the study seeks to contextualize Tunisia's pragmatic, cost-driven adoption pathway through a comparative analysis with Japan, highlighting how technological divergence is a strategic adaptation to local constraints rather than a mere development gap. Ultimately, the analysis provides a framework for understanding context-driven technology adoption to promote data-driven development in resource-constrained economies.

Table 5 Possible applications cases in multiple Tunisian cities.

City	Agriculture	Archeology	Infrastructure
Tunis	- AI-driven crop analysis (e.g., Smart Farm's partnership with Farmitry for real-time soil and irrigation optimization). - Dairy monitoring via AI-collars (Moome Startup tracks cow health, reproduction, and milk quality). - Precision farming platforms leveraging multispectral UAV imagery for pest detection and yield prediction.	- Limited direct applications; however, the digital economy supports AI tools for heritage documentation (e.g., 3D modeling of historical sites like the Medina).	- Urban planning using UAV surveys for traffic management and infrastructure mapping (e.g., Smart City initiatives). - Port logistics enhanced by AI for cargo optimization (Sfax's model may influence Tunis).
Sfax	- Renewable energy integration with UAVs inspecting solar farms for panel defects (thermal imaging + AI analysis). - Automotive parts supply chain optimized via AI-driven logistics (e.g., just-in-time manufacturing).	- Potential for UAV surveys of ancient Punic/Roman sites near the city (e.g., Kerkouane), though no explicit projects found.	- Solar farm maintenance using UAVs for automated defect detection (aligned with Tunisia's 2035 green energy goals). - Port operations (e.g., container traffic management) may use AI for route optimization.
Sousse	- Tourism-driven agriculture (e.g., olive oil exports) supported by UAVs for crop health monitoring in small farms.	- Cultural heritage preservation via UAV surveys of Roman ruins (e.g., Amphitheater of El Jem) for structural analysis.	- Port infrastructure (e.g., cargo handling) may adopt AI for efficiency, though details are sparse. - Resort management using UAVs for beach surveillance and crowd control.
Kairouan	- AI for water-scarce agriculture (e.g., date palm irrigation optimization).- Handicrafts sector may use AI for market demand forecasting (e.g., pottery exports).	- Religious heritage monitoring via UAVs for mosque structural inspections (e.g., Great Mosque of Kairouan).	- Local infrastructure maintenance (e.g., roads) using UAVs for damage assessment, though limited data exists.
Nabeul	- UAV-based precision agriculture (RoboCare uses multispectral drones to reduce water/fertilizer use by 30% in citrus orchards).- AI for crop disease detection (e.g., early identification of citrus greening).	- Pottery heritage digitization using UAVs for 3D modeling of workshops and kilns.	- Coastal infrastructure (e.g., erosion monitoring) via UAV surveys.
Gabès	- Fisheries management using AI to predict fish stocks and optimize catch sizes (EU-certified sardine exports).	- Underwater archeology potential (e.g., shipwrecks in the Gulf of Gabès) using UAVs for preliminary surveys.	- Phosphate mining infrastructure (e.g., railroads) monitored via UAVs for maintenance (though AI integration unclear).- Renewable energy projects (e.g., solar farms) using UAVs for inspections.
Bizerte	- Limited agricultural focus; economy centered on heavy industry.	- No explicit projects, but nearby historical sites (e.g., Carthaginian ruins) could benefit from UAV surveys.	- Bridge construction (Chinese-funded project) using UAVs for progress monitoring and structural analysis (e.g., Bizerte Canal Bridge). - Shipyard maintenance using UAVs for hull inspections (e.g., CMR Tunisia Ship Repairs).
Gafsa	- Phosphate mining's environmental impact monitored via UAVs for soil degradation assessment (though AI integration unclear). - Date palm agriculture supported by AI for irrigation in arid zones.	- No explicit projects, but ancient Roman sites (e.g., Thugga) near Gafsa could use UAVs for mapping.	- Phosphate mining infrastructure (e.g., conveyor belts) may use UAVs for maintenance, though AI applications are speculative.
Mahdia	- Tourism-linked agriculture (e.g., fishing) supported by UAVs for coastal resource monitoring.	- Underwater archeology (e.g., 10th-century shipwreck) using UAVs for preliminary surveys and 3D modeling (Mahdia Museum collaboration).	- Coastal infrastructure (e.g., erosion control) via UAV surveys.
Kasserine	- AI-driven olive oil production (e.g., yield prediction and pest detection). - Solar-powered UAVs for crop monitoring in water-scarce regions.	- No explicit projects, but nearby historical sites (e.g., Roman forts) could benefit from UAV surveys.	- Local roads and irrigation canals monitored via UAVs for damage assessment.
Béja	- Agriculture optimization using UAVs for soil analysis and crop health in wheat/citrus farms.	- No explicit projects, but Roman ruins (e.g., Ammaedara) near Béja could use UAVs for mapping.	- Limited infrastructure data; UAVs may be used for local road maintenance.
Le Kef	- Phosphate mining's environmental impact monitored via UAVs for soil erosion.	- No explicit projects, but historical sites (e.g., Numidian tombs) could benefit from UAV surveys.	- Local infrastructure (e.g., phosphate transport roads) monitored via UAVs for damage assessment.

3. STUDY SITE AND DEVICES

(1) Study site

As shown in **Table 5** and **Fig.2**, Tunisia stands at the intersection of traditional agriculture, archaeological heritage, infrastructure maintenance and AI-driven innovation, leveraging UAVs to address water scarcity and climate challenges. The Ramli agricultural system¹⁹⁾ in Ghar El Melh, a UNESCO-recognized Globally Important Agricultural Heritage System (GIAHS), exemplifies centuries-old resilience through passive irrigation

using tidal movements to sustain crops on sand. Meanwhile, AI and UAVs are revolutionizing precision farming: startups like RoboCare use multispectral imaging and spectral analysis to optimize irrigation, reduce water consumption, and target fertilizers, addressing drought and soil degradation. Archaeologically, Tunisia's Roman and Islamic heritage sites, such as the ancient city of Utica and the Roman amphitheater in El Jem, can benefit from LiDAR and satellite remote sensing for non-invasive mapping and preservation.

Government initiatives, like the SfaxForward



Fig.2 Tunisia mapping of Drones and AI applications (generated by DouBao AI, inspired by human ideas).

project, integrate cultural heritage with sustainable development, fostering inclusive growth through interdisciplinary research and community engagement . This synergy of ancient knowledge,

cutting-edge technology, and policy support positions Tunisia as a model for balancing heritage conservation with climate-resilient agriculture in arid regions.



Fig.3 Drones diagrams and illustrations (from up to down, Mavic, Phantom, Matrice, AGRAS T). Figures were generated by DouBao AI, inspired by human ideas and DJI manuals.

(2) Device

As shown in **Fig.3**, the operational backbone of Tunisia's drone applications relies on a strategic selection of UAVs), primarily comprising multirotor and fixed-wing models chosen for their complementary capabilities. Multirotor drones, such as the agile DJI Phantom 4 RTK²⁰) and the robust DJI Matrice 300 RTK²¹), are favored for their stability and precision in low-altitude missions, ideal for detailed crop inspection or close-range infrastructure monitoring. In contrast, fixed-wing UAVs like the senseFly eBee X excel in covering vast expanses, such as archaeological sites or large farms, due to their longer flight endurance. A critical feature unifying these platforms is their support for a versatile array of sensors – including RGB, multispectral, thermal, and LiDAR cameras – enabling a single device to be adapted for diverse tasks, from analyzing plant health to mapping subterranean ruins. This flexibility, combined with lightweight designs and user-friendly operation, makes these drones particularly well-suited to the cost-effective and mission-specific demands of the

Tunisian context.

Central to the data-capture capabilities of Tunisian drones are multispectral sensors, with the Red Edge sensor being a prominent example. Unlike standard RGB cameras that capture only visible light, multispectral sensors detect specific wavelengths of light, both within and beyond the visible spectrum—including near-infrared (NIR) and red-edge light, which is the narrow band of light between red and NIR where plants reflect light most dramatically when healthy. By measuring the intensity of light reflected in these discrete bands, these sensors reveal critical information invisible to the naked eye. This data is processed into indices like the Normalized Difference Vegetation Index (NDVI), which provides a quantitative measure of plant health, hydration levels, and chlorophyll content. In the Tunisian agricultural context, this translates into the ability to precisely identify areas of crop stress due to water scarcity, nutrient deficiency, or pest infestation long before visible symptoms appear, enabling targeted interventions that optimize resource use and bolster sustainability.

4. METHODOLOGY

This study employs a multi-sector application review, synthesizing data from a multi-source analysis to investigate current and future synergy of AI and drones across Tunisian sectors. The methodology is structured in three primary phases:

Literature and Document Review: A systematic review of academic publications, technical reports from international organizations (e.g., FAO, UNESCO), and policy documents from Tunisian governmental bodies was conducted. This provided the foundational context on national priorities, technological adoption levels, and existing projects in agriculture, archaeology, and infrastructure.

Case Study Identification and Analysis: Specific, documented applications of UAV and AI technology in Tunisia were identified through the literature and public domain records from startups, research institutions (e.g., University of Carthage), and development projects (e.g., SfaxForward). Each case was analyzed to extract key parameters, including: the purpose of drone deployment, the type of UAV and sensors used, the specific role of AI in data processing, and the reported outcomes or potential benefits.

Comparative Analysis: The technological application and maturity in Tunisia were contrasted with a global leader in the field, Japan. This comparison was framed along key dimensions: technological sophistication, primary application goals (e.g., automation vs. monitoring), and adaptation to local socio-economic and environmental constraints. The insights from this phase helped contextualize Tunisia's position in the global landscape and identify potential pathways for future development.

The technological divergence between Japan and Tunisia is not merely a gap in capability but a direct reflection of their underlying socio-economic imperatives. Japan's drive toward full automation is a strategic response to its acute aging population and high labor costs, making capital-intensive technology essential for maintaining operations. In contrast, Tunisia's focus on monitoring and control stems from an existential crisis of water scarcity and stringent budget constraints, prioritizing technologies that maximize conservation and optimize a scarce resource. Thus, their differing approaches—one towards autonomous systems, the other towards efficient management—reveal how national priorities are shaped by distinct demographic, economic, and environmental pressures.

The data synthesized from these phases were used

to populate the analytical tables (Tables 1-5) and to generate the supporting infographic and map (Fig. 1-3), providing a visual summary of the findings.

5. RESULTS AND ANALYSIS

The analysis of the identified cases reveals a clear pattern of targeted, pragmatic application of AI-drone synergy in Tunisia, with distinct characteristics in each sector.

(1) Agricultural Sector: Precision for Sustainability

The results confirm that the primary driver for AI-UAV adoption in Tunisian agriculture is the pressing need to enhance water efficiency and crop resilience. As illustrated in the Ghar El Melh case, the integration is not merely technological but contextual. Drones equipped with multispectral sensors (e.g., Red Edge) capture data that is processed by AI algorithms to generate NDVI and other vegetation indices. The key outcome is the transition from blanket resource application to targeted interventions. For instance, AI-powered analysis can pinpoint specific areas of a field suffering from water stress, allowing farmers to irrigate precisely, thereby potentially reducing water usage compared to traditional methods. This data-driven approach directly addresses the challenges of small farm sizes and labor shortages by providing smallholder farmers with actionable insights that were previously accessible only to large agribusinesses.

(2) Archaeological Sector: Digital Preservation and Discovery

In archaeology, the synergy has proven transformative for heritage management. The application of LiDAR and high-resolution photogrammetry via fixed-wing drones (e.g., senseFly eBee X) over sites like Utica has enabled the creation of sub-meter accuracy 3D models. The AI component is critical in analyzing this data; machine learning algorithms can be trained to identify subtle topographic anomalies indicative of buried structures, effectively "seeing" through surface vegetation. This non-invasive method accelerates survey work and provides a permanent digital record of sites threatened by erosion and urbanization. The results demonstrate a significant reduction in the time and cost required for initial site assessment and ongoing monitoring, enhancing the capacity of Tunisia's limited archaeological conservation resources.

(3) Infrastructure Sector: Enhancing Safety and Predictive Maintenance

For infrastructure, the results highlight safety and efficiency gains. The use of multirotor drones like the DJI Matrice 300 for inspecting bridges or power lines eliminates the need for inspectors to work at height or in hazardous environments. The integration of AI moves the application beyond simple visual documentation towards predictive maintenance. For example, AI algorithms analyzing sequential thermal imagery of electrical grids can detect evolving heat patterns that predict transformer failures before they occur. This shift from reactive repair to predictive maintenance, as facilitated by the AI-UAV combo, is crucial for extending the lifespan of critical national infrastructure and ensuring service reliability.

(4) Summary

A cross-analysis of the results, as summarized in Table 5, reveals several unifying themes:

Technology Appropriation: Tunisia consistently adopts robust, commercially available platforms (predominantly from DJI) rather than developing custom solutions. This suggests a strategy focused on cost-effectiveness and rapid deployment.

AI's Role as a Force Multiplier: Across all sectors, AI's primary value is in automating the analysis of large datasets collected by drones, turning raw imagery into actionable intelligence and thus overcoming human capacity limitations.

Contextual Adaptation: The applications are finely tuned to local challenges: water scarcity in agriculture, site preservation in archaeology, and budget constraints in infrastructure. This contrasts with Japan's focus on high-end automation and efficiency.

6. DISCUSSION

(1) A Catalyst for Data-Driven Development

The fusion of UAV and AI technologies is acting as a catalyst for a smarter, more data-driven approach to national development. In agriculture, it empowers a shift from subsistence-based intuition to precision farming, enhancing climate resilience. For archaeology, it facilitates a leap from manual, risky documentation to digital conservation, safeguarding cultural capital. In infrastructure, it promotes a transition from costly, reactive maintenance to proactive, condition-based management. This synergy effectively democratizes advanced data analytics, making them accessible for solving Tunisia's unique, on-the-ground challenges.

(2) Navigating the Challenges

Despite the promising results, significant challenges must be navigated to fully realize the benefits of this technological synergy.

Regulatory Frameworks: Tunisia's regulatory environment for UAVs is still evolving. Clear, streamlined policies regarding flight permissions, privacy, and airspace integration are essential for widespread adoption.

Technical Capacity Building: The effective use of AI-drone systems requires specialized skills in data processing, GIS analysis, and machine learning. There is a pressing need for targeted educational programs and vocational training to build a local talent pool and prevent a reliance on foreign expertise.

Data Management and Infrastructure: The volume of data generated by these systems necessitates robust data management plans, including storage, processing power, and cybersecurity measures. The lack of adequate digital infrastructure could become a bottleneck.

Economic Accessibility: The initial investment for advanced drones and AI software can be prohibitive for small-scale farmers or local archaeological teams. Exploring innovative business models, such as drone-as-a-service (DaaS) offerings from startups like RoboCare, is critical for improving accessibility.

7. CONCLUSION

This analysis confirms the transformative role of AI-UAV synergy as a strategic tool for Tunisia's development. The multi-sector investigation demonstrates that this integration directly addresses critical national challenges: enhancing water efficiency in agriculture, enabling non-invasive digital preservation of cultural heritage, and improving the safety and predictive maintenance of infrastructure. The core finding is that Tunisia's technological pathway is not a lagging imitation but a deliberate contextual appropriation. The divergence from Japan's automation-focused model—prioritizing cost-effective monitoring over high-end automation—is a rational adaptation to local constraints like water scarcity and budget limitations. This pragmatic approach offers a replicable framework for tailoring technology to resource-constrained economies. Ultimately, solidifying this potential hinges on overcoming key barriers: establishing clear regulations, building local technical capacity, and ensuring economic accessibility. By strategically navigating these challenges, Tunisia can leverage this synergy to build a more resilient and data-driven future.

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ADDITIONAL MATERIAL

(1) Purposes of Using Drones in Tunisia

As shown in **Table 1**, the main objective of drone implementation in Tunisia focuses on improving agricultural efficiency and sustainability through precision farming methods¹¹⁾. The agricultural sector utilizes drones to conduct crop inspections and soil evaluations and manage irrigation systems and measure crop production levels. The technology enables farmers to maximize their water and fertilizer resources while enabling disease detection at an early stage and decreasing their need for manual work. UAVs deliver essential data to help decision makers enhance resource management in areas experiencing water shortages and climate-related challenges^{11), 12)}.

Beyond their core role in agriculture, drones are increasingly applied to address critical needs in Tunisia's archeology and infrastructure maintenance sectors, further demonstrating their versatility in tackling the country's unique challenges. In archeology, a field central to preserving Tunisia's rich historical heritage (including UNESCO sites like Carthage and Dougga), drones overcome longstanding hurdles such as the vastness of archaeological landscapes and the risk of damaging fragile ruins during ground exploration.

Equipped with high-resolution RGB cameras, thermal sensors, and LiDAR technology—similar to their agricultural use but calibrated for historical contexts—UAVs capture detailed aerial imagery that reveals hidden structures (like ancient city walls or burial sites) beneath dense vegetation and identifies soil temperature variations that signal underground artifacts. This data is used to create precise 3D models of sites, enabling non-invasive mapping, analysis, and long-term monitoring to protect heritage from erosion, urbanization, or unauthorized activity.

Meanwhile, in infrastructure maintenance—critical for supporting Tunisia's economy and daily life—drones address the inefficiencies, safety hazards, and accessibility issues of traditional methods. For key structures like bridges, highways, power lines, and water pipelines, UAVs with high-resolution sensors and thermal imaging detect defects such as bridge cracks, power line corrosion, or pipeline leaks early on; LiDAR even generates 3D models of highways to map pavement deterioration. By reducing the need

for workers to access high-risk areas (e.g., tall bridges or high-voltage towers) and cutting inspection time and costs, drones not only improve operational safety but also extend the lifespan of infrastructure—helping mitigate the impact of limited maintenance budgets and ensuring reliable services across the country.

(2) Types of Drones Used in Tunisia

As shown in **Table 2**, the research institutions and private companies in Tunisia operate multirotor and fixed-wing UAVs based on their specific mission requirements. The drone models used in Tunisia include DJI Mavic, DJI Phantom, DJI Matrice, DJI AGRAS T and SenseFly eBee X^{13), 14)}. The drones used in Tunisia possess lightweight construction and simple operation features while supporting various sensor types.

In Tunisia, the selection of drones is tightly tied to the unique demands of each mission, with research institutions and private companies relying on two primary categories: multirotor and fixed-wing UAVs, each optimized for different tasks. Multirotor drones, exemplified by the widely used DJI drones, excel in low-altitude, precise operations—thanks to their lightweight construction and simple maneuverability.

These models are ideal for agricultural tasks like crop mapping and soil analysis, as they can hover steadily to capture high-resolution data with their integrated RGB and multispectral sensors, and their 30-minute flight time balances efficiency with detailed coverage. Fixed-wing drones, such as the senseFly eBee X, by contrast, are designed for large-area surveys, offering longer flight endurance and greater range than multirotors—making them well-suited for archaeological site mapping or wide-scale infrastructure inspections.

A key unifying feature across both types is their compatibility with a diverse suite of sensors, including RGB, multispectral, thermal, and LiDAR cameras. This flexibility allows the same drone platform to adapt to multiple use cases: for example, a multirotor can switch from using RGB sensors for crop health checks to thermal sensors for detecting irrigation leaks, while a fixed-wing drone can leverage LiDAR to penetrate vegetation and map hidden archaeological ruins.

By combining lightweight builds, user-friendly operation, and broad sensor support, these drones—whether multirotor or fixed-wing—ensure that Tunisian organizations can address sector-specific needs, from precision agriculture to heritage preservation and infrastructure monitoring, with tailored, efficient technology.

(3) The implementation of AI technology with UAV systems in Tunisia requires explanation

As shown in **Table 3**, AI technology improves UAV operations through its ability to perform automated data processing. The system uses AI algorithms to analyze aerial images which results in automatic detection of crop diseases and weed identification and yield prediction and soil classification¹⁵. The combination of AI with UAV systems in Tunisia enables solutions for three major agricultural problems which include small farm sizes and insufficient workers and insufficient water resources. AI + UAV systems generate immediate data which enables farmers to make quicker and more intelligent agricultural choices¹⁶.

The implementation of AI technology alongside UAV systems in Tunisia is a strategic response to turning raw drone-captured data into actionable insights, directly addressing the country's most pressing agricultural pain points. While UAVs excel at collecting high-volume data – from RGB and multispectral imagery of crops to soil condition scans – AI fills the critical role of processing this data automatically and accurately, eliminating the delays and human error of manual analysis. AI algorithms are trained to interpret aerial data for targeted agricultural tasks: they can distinguish subtle visual cues to detect early-stage crop diseases (before symptoms are visible to the naked eye), identify invasive weeds that compete for resources, predict crop yields based on growth patterns, and classify soil types by analyzing spectral data—all in a fraction of the time it would take human teams.

This integration is particularly impactful for Tunisia's agricultural context, where small farm sizes limit economies of scale, labor shortages strain manual monitoring efforts, and water scarcity demands precise resource use. For example, an AI-UAV system can quickly map a smallholder's field, flag a patch of diseased wheat, and recommend a targeted pesticide application—avoiding the waste of treating the entire field and reducing chemical use. By generating immediate, hyper-localized data, AI-UAV systems empower Tunisian farmers to move beyond reactive decision-making (e.g., addressing a full-blown disease outbreak) to proactive, precision-driven choices – whether optimizing irrigation schedules for water-scarce regions or adjusting fertilizer use to match soil needs. In turn, this not only boosts individual farm productivity but also aligns with Tunisia's broader goal of building more sustainable, resilient agricultural systems amid climate variability.

(4) Comparison Between Tunisia and Japan in AI-Based UAV Applications

As shown in **Table 4**, The analysis examines how Tunisia and Japan implement AI-based UAV systems in their operations. Japan stands as a worldwide leader in smart agriculture but Tunisia exists at the beginning of adopting this technology¹⁷. The Japanese agricultural sector employs UAVs with AI-powered systems and advanced sensors to perform automated spraying operations and precision mapping tasks¹⁸. The main drone applications in Tunisia involve electric drones which perform monitoring and data collection tasks while emphasizing cost-effectiveness and operational flexibility in dry environments¹⁶.

A comparison of Tunisia and Japan in AI-based UAV applications reveals contrasts rooted in technological maturity, operational goals, and adaptation to local agricultural realities. Japan, a global pioneer in smart agriculture, leverages its advanced R&D capabilities to integrate AI-UAV systems into end-to-end workflows prioritizing efficiency and cutting-edge functionality.

Its UAVs are often equipped with AI-powered systems and high-end sensors, enabling tasks that go beyond data collection: AI-driven automated spraying, where algorithms calculate precise pesticide or fertilizer doses based on real-time crop health data, and advanced precision mapping that generates centimeter-accurate field models for long-term yield optimization. These systems are tailored to Japan agricultural landscape – characterized by large, mechanized farms and a focus on minimizing labor. By contrast, Tunisia, which is in the early stages of adopting AI-UAV technology, designs its approach around practicality, cost-effectiveness, and resilience to its arid climate.

It relies primarily on electric drones that focus on core AI-enhanced tasks: monitoring crop health, collecting soil data, and detecting early-stage diseases or irrigation gaps. These operations prioritize flexibility—electric drones can be deployed quickly across Tunisia small, fragmented farms—and adaptability to dry environments, where water scarcity demands targeted resource use over large-scale automation. While Japan AI-UAV aims to push the boundaries of smart agriculture, Tunisia focus is on incremental, accessible solutions that address its most urgent challenges: labor shortages, small farm sizes, and water constraints. This difference reflects not just a gap in technological advancement, but also how each country aligns AI-UAV tools with its unique agricultural needs—Japan optimizing for scale and automation, Tunisia for affordability and climate resilience.

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